

Mapping food-logistics nodes and adaptive supply corridors under blockade: a geospatial analysis of Myanmar's western (Kaladan-corridor) borderland

Abstract

Armed conflict disrupts food access not only by damaging production, but by controlling the corridors, crossings, markets and humanitarian channels through which food moves. Yet the spatial mechanisms through which blockade becomes localised hunger remain poorly mapped. This paper develops a node–corridor–chokepoint framework for analysing conflict-sensitive food logistics and applies it to Myanmar's western borderland, defined as a functional supply system linking the food-deficit Chin highlands with the Kaladan riverine corridor, the dry-zone gateway belt and the Mizoram crossing. Using local-news reporting, ACLED conflict records, OpenStreetMap transport data and SPAM 2020 crop-production grids, we reconstruct a reported logistics-association network and analyse it through centrality, corridor viability and node-removal simulations. Three findings emerge. First, logistics-relevant disruption rises sharply after the 2021 coup and persists through 2025–2026. Second, the regional system resolves into a triad: Gangaw/Yaw as a contested production source, Rikhawdar–Mizoram as a terminal cross-border gateway and Paletwa–Kaladan as the strategic chokepoint. Third, the network is robust to random loss but highly vulnerable to targeted removal: removing the top 5% of nodes by betweenness reduces the largest connected component to 7%. The study shows how conflict-induced food insecurity can be diagnosed as a spatial-circulation problem and identifies leverage points for humanitarian logistics in contested borderlands.

Keywords: Food access; conflict; humanitarian logistics; logistics chokepoints; borderlands; spatial network analysis

1. Introduction

Armed conflict reshapes human security not only through the violence it inflicts directly, but through the slow disruption of the everyday circulation systems on which survival depends. Food logistics—the movement of staple commodities across production areas, markets, border crossings and the last miles into the household—is among the most consequential of these systems, and among the least mapped under wartime conditions. Contemporary wars are fought as much over the contestation of transport corridors, the interdiction of convoys, the destruction of bridges and the administrative throttling of cross-border trade as over ground itself, and the cumulative effect is a *de facto* blockade that converts geopolitical violence into localised hunger. International law has begun to register the gravity of this: the United Nations Security Council has condemned the use of starvation and the unlawful denial of humanitarian access, warning that to starve civilians as a method of warfare may amount to a war crime (UNSC

2018; Grace 2025).

Myanmar since the 2021 coup is an acute instance. The military's long-standing 'four cuts' doctrine—severing food, funds, intelligence and recruits from areas it cannot hold—has been generalised to national scale, with trunk routes and waterways blocked and humanitarian actors barred from reaching millions in need (OHCHR 2023). By early 2025 roughly a third of the population was food insecure, internal displacement had risen by 72% in a single year to 3.5 million, and Rakhine State stood on the brink of famine with some two million people at risk; United Nations experts judged the near-total blockade of aid to Rakhine since 2023 a likely war crime (OHCHR 2025; WFP 2023). The crisis has, moreover, been actively obscured, with investigative reporting documenting the suppression of food-security analyses that pointed to one of the world's worst contemporary food emergencies (BROUK 2025).

What remains underdetermined, despite this mounting concern, is the spatial anatomy of the blockade itself: which nodes concentrate supply dependence, which corridors retain viability, and how provisioning reroutes once a primary link is severed. Section 2 assembles four strands of scholarship—on conflict and food access, on critical logistics and chokepoints, on spatial networks and resilience, and on the computational extraction of conflict events—and draws them into an analytical framework that Sections 3 to 5 then operationalise for the western (Kaladan-corridor) borderland. Three questions organise the inquiry. **(RQ1)** Where are the critical food-logistics nodes that structure food access under blockade? **(RQ2)** How do conflict disruptions reconfigure corridor viability across rugged terrain and cross-border interfaces, and what adaptive pathways emerge? **(RQ3)** Which nodes and corridors carry the highest systemic vulnerability, and what patterns of redundancy point to resilience? In answering them the paper supplies both a transferable method for locating leverage points in conflict-affected circulation systems and a geographically grounded reading of one of Asia's most acute food emergencies.

2. Conceptual background and related work

2.1 Conflict, blockade and food access

Hunger in war is, more often than not, produced rather than simply suffered. The political economy of famine has long shown that the denial of food and of movement can be a deliberate instrument of control and accumulation rather than a by-product of fighting (Keen 1994; de Waal 2017), and that even relief can be captured, taxed or redirected by the armed actors it is meant to bypass (Nunn and Qian 2014). More recent work establishes that the obstruction of humanitarian access—worked through checkpoints, movement restrictions, harassment and the

targeting of convoys—is not an aberration but a recurring strategy that determines who can be reached, and when (Grace 2025), and that this strategy is now recognised in international law as a potential war crime (UNSC 2018). In Myanmar the logic is doctrinal, codified in the ‘four cuts’ and generalised after the coup (OHCHR 2023), with famine-level risk as its consequence (OHCHR 2025; WFP 2023). What this literature documents, however, is largely the incidence and the humanitarian toll of blockade; the spatial mechanics by which a food-supply system is actually severed, and through which it reorganises, remain comparatively unexamined.

2.2 Critical logistics and chokepoint geographies

A second strand reframes circulation itself as a site of power. Critical-logistics scholarship argues that control over the movement of goods—securing some flows while interrupting others—has become central to contemporary statecraft and violence (Cowen 2014). Building on this, work on chokepoints and logistical bottlenecks shows that concentrated control over a non-substitutable passage confers leverage disproportionate to the territory physically held, making such points both strategic assets and sites of resistance (Danyluk 2023; Dittmer 2021). Empirically, transport networks have been shown to structure the diffusion of insurgent violence, with roads and routes shaping where conflict spreads (Zhukov 2012). Yet this literature is largely qualitative or focused on military and territorial dynamics; it rarely reconstructs the civilian food-provisioning network or quantifies which nodes hold the greatest systemic leverage.

2.3 Spatial networks, centrality and resilience

Network science provides principled tools for exactly this. Betweenness centrality measures the extent to which a node lies on shortest paths between others, identifying intermediaries whose removal would most disrupt flow (Freeman 1977); the interpretation of centrality as control over network flow is well established (Borgatti 2005). Robustness analysis further shows that heterogeneous networks are typically tolerant of random failure but acutely fragile to the targeted removal of high-centrality nodes—the canonical “robust-yet-fragile”, single-point-of-failure result (Albert et al. 2000). Complementary resilience concepts frame the capacity of a system to maintain function through redundancy, rerouting and adaptive reorganisation when components are lost (Holling 1973; Cumming 2011). These tools are widely applied to transport and infrastructure systems, but seldom to conflict-disrupted food logistics with empirically reconstructed nodes and corridors—leaving the systemic importance and failure consequences of specific supply nodes largely unquantified.

2.4 Detecting disruption from unstructured text

A fourth strand concerns how disruption is observed. Automatically coded global event datasets such as GDELT offer scale but have well-documented accuracy problems in subnational conflict settings (Wang et al. 2016; Yuan et al. 2017), while curated datasets such as ACLED provide reliable, geolocated events (Raleigh et al. 2010). Advances in named-entity recognition, geoparsing and, more recently, large-language-model relation extraction make it increasingly feasible to convert place-based disruption narratives in news into structured, geocoded events, supplying the timing, causation and local detail that gridded or pre-coded sources lack (Chen et al. 2021; Leppamäki et al. 2024; Mai et al. 2024). Such methods nonetheless inherit reporting and linguistic biases, and language models themselves can be geographically biased, so extraction must be triangulated and interpreted with care (Baum and Zhukov 2015; Shaver et al. 2022; Manvi et al. 2024).

2.5 Research gap and analytical framework

Taken together, these strands document the humanitarian outcomes of conflict-induced hunger, theorise power over circulation and the chokepoint, furnish the network and resilience tools through which flows can be analysed, and make disruption legible from text; what none of them does is to integrate these moves into a spatially explicit account that reconstructs a conflict-affected food-logistics network, quantifies the systemic weight of its nodes, and stress-tests its vulnerability. The framework developed here is built to close that gap, and it proceeds in two moves. The first represents food provisioning as a spatial network of logistics nodes—production hinterlands, market towns, transshipment points, and cross-border or riverine gateways—joined by supply corridors, the roads, river channels, mountain paths and multimodal routes along which commodities move under terrain and risk constraints. The second specifies how that network is to be read empirically, assigning to each component a characteristic mode of disruption and a measurable indicator (Figure 1).

Within this network we distinguish three functional node types, each corresponding to a distinct mode of disruption and to a measurable indicator (Figure 1). A production source / transfer interface is where production hinterlands meet the transport network; its contestation degrades source areas. A strategic chokepoint is a node whose control over a non-substitutable corridor confers disproportionate leverage over circulation across a wider territory; its systemic importance is approximated by high betweenness centrality and probed by the impact of its removal. A cross-border / terminal gateway is an external entry point that becomes the binding constraint on inflows when internal corridors fail; it appears as a peripheral, low-betweenness

node whose value is as an alternative endpoint. Corridor viability is assessed through accessibility (travel-cost) surfaces combined with conflict-exposure friction, and overall resilience is quantified by simulating node removal and measuring how quickly connectivity collapses relative to random loss. The framework directs the analysis toward identifying high-centrality nodes (RQ1), characterising corridor viability and adaptive rerouting (RQ2), and stress-testing systemic vulnerability and redundancy (RQ3).

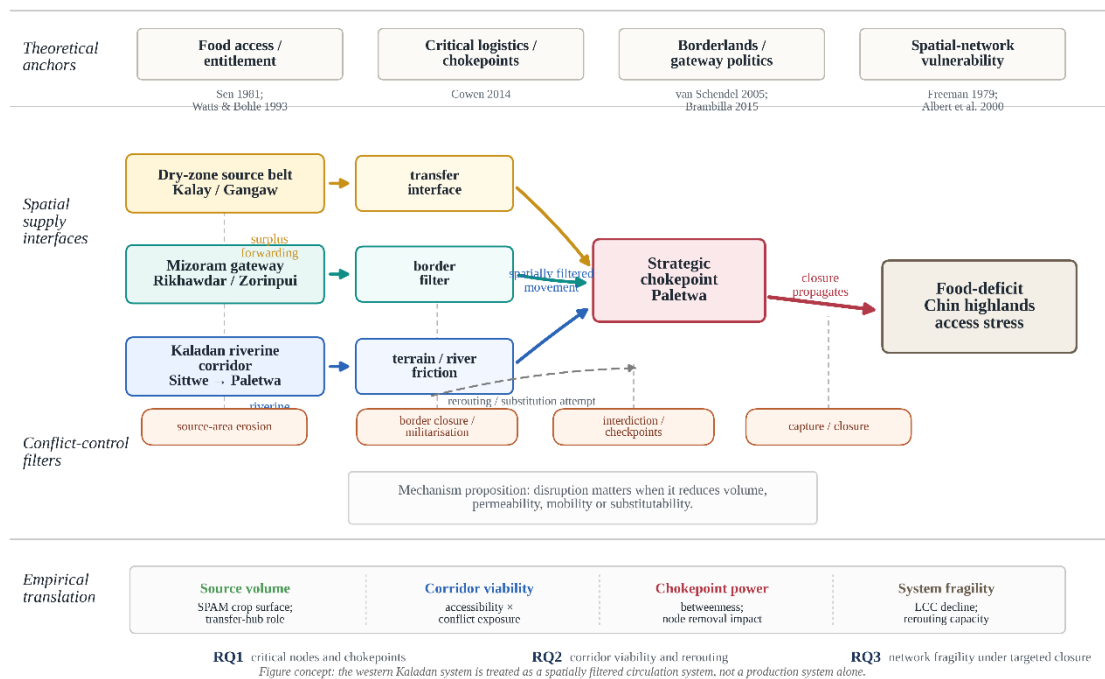


Fig. 1. The node–corridor–chokepoint framework: node types, disruption mechanisms, indicators and research questions.

3. Study area

We operationalise the framework for the functional food-supply system of Myanmar’s western borderland, not a single administrative unit (Fig. 2). Its core is the food-deficit Chin highlands—rugged western uplands with elevations above 3,000 m, deeply incised valleys and few all-weather roads, making the area one of the least accessible and least food-secure parts of the country. Because the highlands produce little of their own staple supply, the relevant system extends to the interfaces that feed them: eastward to the dry-zone gateway belt around Gangaw, the Yaw valley, Kalay and Tamu (in Magway and Sagaing), which channels surplus rice into the hills; southward to the Kaladan riverine corridor (Paletwa–Kyauktaw–Minbya–Sittwe, mostly in Rakhine), the only substantial waterway linking inland Chin and Mizoram to the Bay of Bengal through the India-funded Kaladan Multi-Modal Transit Transport Project; and westward to the Rikhawdar crossing into India’s Mizoram. Defining the region functionally is essential: an administrative “Chin State” frame would be data-thin in the uplands

and would exclude Paletwa, Kyauktaw and Sittwe—the Kaladan nodes that in fact govern southern supply.

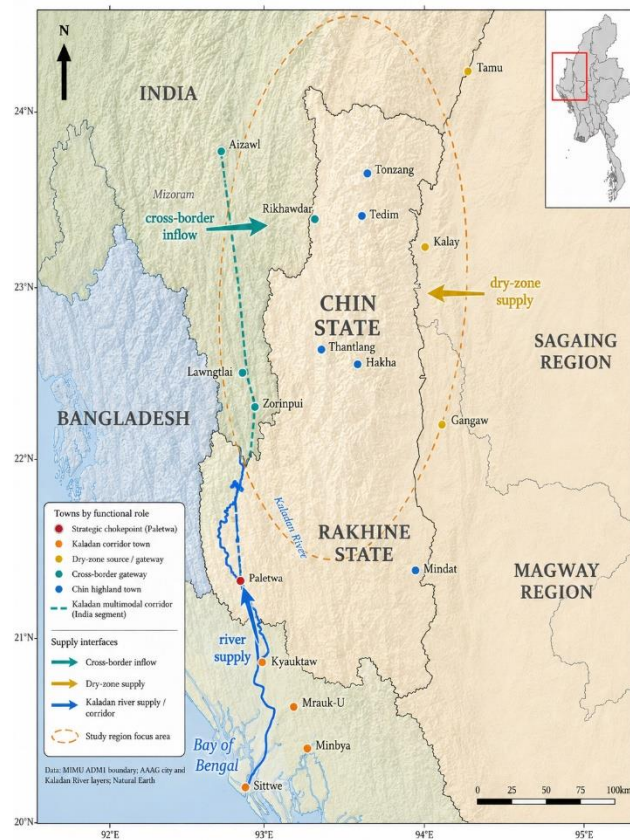


Fig. 2. The western (Kaladan-corridor) food-supply system. Towns are coloured by their functional role in the framework (Table 1); arrows show the three supply interfaces feeding the food-deficit Chin highlands: the dry-zone belt (east), the Rikhawdar–Mizoram crossing (west), and the Kaladan riverine corridor up from the coast at Sittwe to Paletwa (south). Inset: location within Myanmar.

Because few redundant routes exist, the closure of a single road, bridge or crossing can isolate large upland populations—a vulnerability that armed actors actively exploit. Two recent developments sharpen the system’s interfaces. In January 2024 the Arakan Army (AA) captured Paletwa town, seizing the junta’s logistics base for the India and Bangladesh borders; the offensive complicated the US\$484-million Kaladan project and triggered air attacks that displaced over 50,000 people (Radio Free Asia 2024; The Irrawaddy 2024). On the western interface, India suspended the long-standing Free Movement Regime in February 2024 and began fencing the 1,643-km border; the Rikhawdar–Zokhawthar friendship bridge was closed to trade and general movement from May 2024, and by December 2024 crossing required biometric border passes limited to a 10-km zone, even as Mizoram hosted some 40,000 Chin refugees at Zokhawthar (Mizzima 2024; Oxford Border Criminologies 2025). On the eastern interface, the dry-zone gateways around Gangaw and Kalay became contested sites of the post-coup war. The western borderland thus exemplifies how blockade dynamics convert a fragile

physical geography into a humanitarian food-access crisis.

4. Data and methods

4.1 Data sources

The analysis integrates textual and geospatial data for Myanmar and, in detail, its western borderland. The textual corpus is drawn from Burma News International (BNI), an independent aggregator of about 15 ethnic and regional newsrooms—including Chin World, Khonumthung News, Kaladan Press, Narinjara and Mizzima—several of which operate partly from outside Myanmar to preserve editorial independence and provide context-rich, less state-filtered coverage of localised disruption. Rather than relying on the pre-coded event table of GDELT, whose accuracy in subnational conflict settings is limited (Wang et al. 2016; Yuan et al. 2017), we engage directly with the raw articles. The corrected event table used in this revision contains 12,371 geocoded event rows from 12,338 unique article URLs. A reproducibility audit of Date_start identifies 11,643 strict full-date rows in the 2016–2026 analysis window, including 545 records in 2026; because the latest dated record is 21 April 2026, 2026 is treated as a partial year in all temporal plots. The functional western/Kaladan study-region filter identifies 2,481 records and 2,480 unique article URLs. Geospatial layers comprise armed-conflict events from ACLED (Raleigh et al. 2010), settlements and transport networks from OpenStreetMap (Haklay and Weber 2008), food-crop production from SPAM 2020 at 10-km resolution (Yu et al. 2020), administrative boundaries from the Myanmar Information Management Unit, and opium-poppy cultivation density from the UNODC opium survey.

Table 2. News corpus composition and event-table audit (source: BNI-derived event table, strict full-date analysis window 2016–2026; 2026 partial).

Subset	Events / rows	Unique articles
Full event table, all years	12,371	12,338
Rows with parseable Date_start	11,677	—
Strict full-date analysis window (2016–2026; 2026 partial)	11,643	11,611
Year-identifiable analysis window (2016–2026; incomplete month/year strings retained)	12,302	12,270
Functional western/Kaladan study-region subset	2,481	2,480
Corrected Fig. 2 logistics subset (non-exclusive category flags; title/event_type/sub_event only)	913	913
Curated regional network input after geocoding/token cleaning	1,066	—

4.2 Text-to-event pipeline for logistics-relevant disruption

Articles were retrieved and filtered with hierarchical place-name recognition that expanded retrieval to western-borderland toponyms (Chin towns, Kaladan-corridor and gateway places) alongside thematic keywords (“conflict”, “security”, “food”, “border”, “Tatmadaw”), with a disambiguation step removing items referring only to Kachin (a substring artefact) or China. Thematic keywords were expanded using cosine similarity between word vectors to capture related terms (e.g., “violence”, “attack” for “conflict”; “smuggling”, “checkpoint” for “border”):

$$\cos = \frac{\sum x_i \times y_i}{\sqrt{\sum x_i^2} \sqrt{\sum y_i^2}} \quad (1)$$

Named-entity recognition extracted actors, organisations and place names; event-level relations were extracted with a large-language-model reasoning step (with dynamic few-shot retrieval and k-nearest-neighbour association), consistent with recent advances in LLM-based extraction and geoparsing for crisis text (Chen et al. 2021; Leppamäki et al. 2024; Mai et al. 2024). The prompt template and coding guide used to standardise the extraction are provided in Appendix A. The pipeline produced 12,371 geocoded event rows across all years, each with a date, three-level location, an initiator–recipient actor dyad, event_type and sub_event. For the temporal analysis, logistics-relevant categories were re-audited from the derived title, event_type and sub_event fields rather than from the full article body, because full-text keyword matching inflated broad categories such as displacement. Six non-exclusive logistics flags were assigned—food shortage / food stress, road or transport closure, checkpoint/control, border-trade disruption, displacement and aid obstruction / humanitarian access—so that narratives could be linked to nodes and corridors rather than to generic conflict incidence. We acknowledge that LLM-based extraction can carry geographic and linguistic biases (Manvi et al. 2024), addressed in the limitations.

4.3 Extraction validation and data quality

Because the extracted events form the empirical basis of the analysis, we evaluate them through a tiered validation strategy that separates reproducible data-quality auditing, geographic consistency checks and analytical robustness from full article-level classification validation. The first tier is a reproducibility audit of the event table itself. Of the 12,371 extracted rows, 94.4% have a parseable Date_start, 99.5% a non-empty state/region field, 88.5% a non-empty town/township field, 66.6% a non-empty village/place field, 99.5% an initiator,

99.6% a recipient, 100.0% an event_type label and 99.0% a sub_event label. No exact duplicate rows were found when records were compared by URL, date, location, event_type and sub_event. The functional western/Kaladan region was re-filtered with a fixed toponym list matching the study-area definition—Chin highland towns, Kaladan-corridor towns, dry-zone gateways and the Rikhawdar–Zokhawthar interface—yielding 2,481 records. Applying the six non-exclusive logistics flags to title, event_type and sub_event yields 913 logistics-relevant records for the temporal analysis. This audit establishes that the analytical subset is internally consistent and reproducible from the submitted event table.

The second tier is geographic and construct validation. Place names are normalised against the MIMU administrative and place-code gazetteer where possible, so that extracted toponyms can be checked against the township/state hierarchy used in the spatial analysis. ACLED is used as an external reference for the timing, location and broad type of conflict events. This comparison does not validate food-logistics categories directly, because ACLED codes political violence rather than food-access disruption; instead, it tests whether the extracted disruption geography is broadly consistent with independently curated conflict incidence. Aggregate plausibility is further assessed against humanitarian displacement and food-security products from HDX/OCHA and WFP, which provide independent temporal and spatial benchmarks for displacement and food-stress patterns. The third tier is analytical robustness: centrality results are tested against alternative edge-weight thresholds and event-bootstrap resampling. The persistence of the same backbone—especially the repeated top ranking of Paletwa and the dominance of Kaladan and dry-zone towns—indicates that the substantive network interpretation is not an artefact of a single edge-definition choice. We therefore interpret the corpus as an audited LLM-assisted event table and the resulting network as a reported logistics-association network.

4.4 Network analysis, accessibility and robustness simulation

Geocoded events were integrated with the geospatial layers in three steps. First, to identify strategic nodes (RQ1), we built toponym co-occurrence networks in which an edge denotes a co-reported logistics relation—two places are linked when they appear together in the same logistics-relevant event, weighted by the number of such co-reports. This is a co-reporting relation, not a verified physical supply link; The co-reporting network is therefore treated as an event-derived logistics-association network. Its nodes are resolvable settlements, townships, river ports, border crossings or market towns named in logistics-relevant disruption narratives. Its edges are weighted co-reporting relations: two places are linked when they appear in the

same event record or article-level disruption narrative, and the edge weight increases with repeated co-reporting. This definition does not assume that every edge is a directly observed shipment route. Rather, it captures the reported operational association between places during disruption—where food stress, road closure, checkpoint control, displacement, border disruption or aid obstruction is narrated as part of the same spatial event. For this reason, centrality is interpreted as brokerage within the reported disruption geography, not as calibrated traffic volume.

Several cleaning steps are used to reduce spurious connectivity. Generic location tokens such as “villages”, “camps”, “border area” and unresolved administrative descriptors are removed unless a specific place name is present (Girvan and Newman 2002). Repeated article-level mentions are collapsed into weighted edges rather than duplicated as independent links, and the regional network is analysed separately from the national benchmark so that high-frequency national hubs do not swamp the internal structure of the western borderland. Robustness is then assessed in two senses. First, node-removal simulations compare targeted removal by betweenness and degree with random removal, tracking the largest connected component as a measure of fragmentation. Second, sensitivity checks vary the minimum edge weight and resample the event set, asking whether the identity of the backbone changes when weaker or less stable co-reporting ties are excluded. The persistence of Paletwa and the Kaladan towns under these tests gives confidence in the relative ranking of chokepoint nodes, while the absolute centrality values are interpreted cautiously. We computed two networks: a national network (2,793-node giant component) used as a benchmark to position the western borderland within the country’s wider logistics structure, and the internal network of the western borderland itself (1,066 regional logistics events; 735-node giant component, 2,790 edges), which is the analytical focus. Second, to delineate source areas and transfer hubs, SPAM 2020 crop yields were overlaid with the OpenStreetMap (OSM) road network; a KD-tree nearest-neighbour search identified transfer hubs, and a 20-km buffer—one day’s walking range—represented the effective last-mile radius under degraded transport. A relative accessibility (travel-cost) surface combined with annual ACLED conflict exposure was used to assess corridor viability and rerouting (RQ2). Third, to quantify systemic vulnerability and redundancy (RQ3), we ran a network-robustness simulation on the regional network: nodes were removed in three orders—by descending betweenness, by descending degree, and at random (mean of 15 runs)—and the size of the largest connected component was tracked. The gap between targeted and random curves measures the extent to which the regional system

depends on a few chokepoints versus distributed redundancy (Albert et al. 2000).

To test whether the centrality findings depend on this network definition, we ran three sensitivity checks on the regional network. Varying the minimum edge weight (≥ 1 , ≥ 2 and ≥ 3 co-reporting events) left Paletwa as the highest-betweenness node and the Kaladan-corridor towns (Kyauktaw, Minbya, Sittwe, Ponnagyun, Mrauk-U) dominant in every case. Across 150 bootstrap resamples of the events, Paletwa was the top-ranked node in 87% of replicates and within the top three in 99%, with 97% of top-three positions occupied by Kaladan and dry-zone backbone towns. Betweenness was only moderately correlated with degree (Spearman $\rho = 0.46$) and weakly with eigenvector centrality ($\rho = 0.18$), consistent with its interpretation as brokerage rather than mere connectivity. The identity of the backbone is therefore robust to network-definition choices, even though absolute centrality values are not; a transport-constrained variant—filtering edges by OpenStreetMap road and river adjacency—remains a priority extension for future work.

5. Results

5.1 Temporal structure of logistics-relevant disruption

The corrected event table confirms that the corpus contains partial 2026 observations rather than a complete 2026 year. Using the fixed western/Kaladan study-region filter and the derived event fields (title, event_type and sub_event), logistics-relevant events rise sharply from 2019, remain elevated after the February 2021 coup and persist through 2025–2026 (Fig. 3). Displacement dominates the annual event flags, while food stress, road/transport closure, checkpoint/control, border-trade disruption and aid/humanitarian-access events show smaller but coherent post-coup increases. The 2026 decline is not interpreted as a substantive de-escalation because the latest dated record is 21 April 2026.

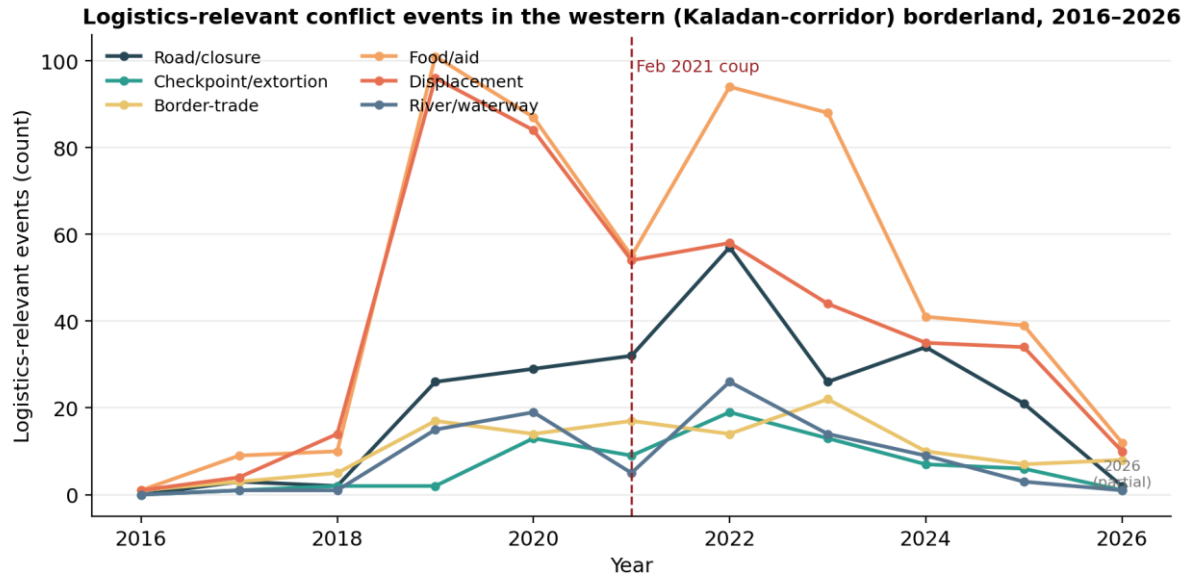


Fig. 3. Temporal structure of logistics-relevant events in the western/Kaladan borderland, 2016–2026, by category. Counts are non-exclusive flags derived from title, event_type and sub_event fields within the fixed 2,481-record study-region subset; 2026 is partial to 21 April. The dashed line marks the February 2021 coup.

5.2 Strategic logistics nodes: national benchmark and regional structure (RQ1)

Betweenness centrality first establishes the national context (Fig. 4a). Nationwide, the highest-betweenness logistics nodes are overwhelmingly border gates and river ports—Myawaddy and Muse (the Thai and Chinese trade gates), Myitkyina and Hpakant (Kachin), and the Rakhine–Kaladan cluster of Ann, Maungdaw, Kyauktaw, Kalay, Kyaukphyu and Minbya—indicating that control over wartime circulation tends to concentrate at a small set of crossings and ports. Paletwa ranks 20th of 2,793 nodes (top 1%) nationally, supporting the external validity of the case.

Within the western borderland itself, the internal network reveals a clear backbone (Fig. 4b). Paletwa ranks first by betweenness (0.261), followed by the Kaladan-corridor towns of Kyauktaw (0.254), Minbya (0.196), Sittwe (0.160), Ponnagyun (0.137) and Mrauk-U (0.100), and by the dry-zone gateway of Kalay (0.231). The Chin highland towns—Hakha (0.064), Falam, Tedim, Thantlang—are comparatively peripheral, consistent with their role as deficit endpoints rather than brokers. Two further findings refine the framework. Gangaw/Yaw, despite its agricultural importance, has low betweenness: it is a production source rather than a through-broker, so its vulnerability lies in source-area degradation. Rikhawdar–Mizoram likewise has low betweenness, consistent with a terminal gateway whose value is as an alternative endpoint. The triad therefore comprises functionally distinct nodes—a high-centrality chokepoint (Paletwa and the Kaladan corridor), a production source area

(Gangaw/Yaw and the dry-zone belt) and a terminal cross-border gateway (Rikhawdar)—each impairing food access through a different mechanism.

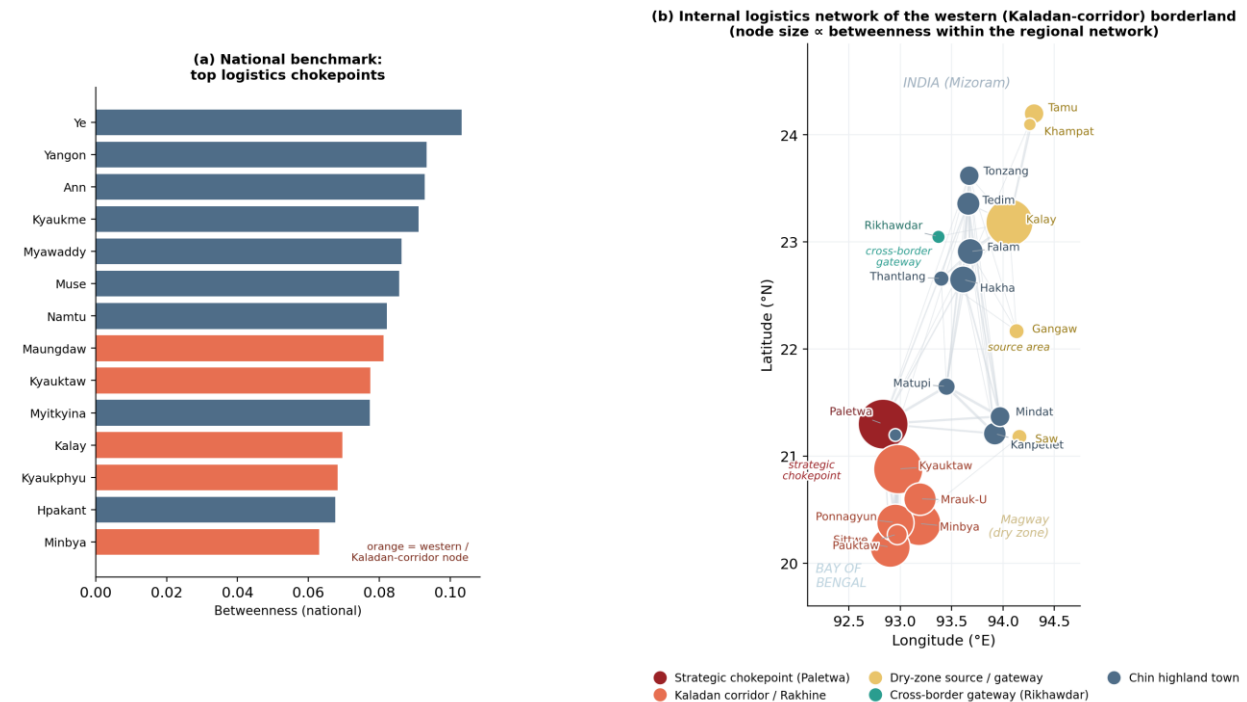


Fig. 4. Logistics chokepoints in the western borderland against a national benchmark. (a) Top logistics nodes nationwide by betweenness; western/Kaladan-corridor nodes in orange. (b) Internal node-link network of the western borderland on geographic coordinates; node size is proportional to betweenness within the regional network. Paletwa anchors the Kaladan corridor (Kyauktaw–Minbya–Sittwe) as the strategic chokepoint; Gangaw and Kalay form the dry-zone source/gateway; Rikhawdar is the cross-border gateway; Chin highland towns are peripheral.

5.3 A triadic logistics structure

5.3.1 Gangaw/Yaw and the dry-zone belt: the production source area

The SPAM 2020 surface shows food-crop production concentrated along the eastern margin of the region—around Kalay, the Yaw valley and Gangaw—rather than in the high interior (Fig. 5b). The KD-tree analysis identifies Gangaw/Yaw as the principal transfer point where dry-zone surplus meets the road network climbing into the highlands, and the one-day-walk buffer captures the last-mile radius under degraded transport. Although its event-network betweenness is low—because it is a source rather than a broker—its strategic value is precisely as the highlands’ food source: after 2021, confrontations around Gangaw and Kalay intensified as resistance forces and the Tatmadaw competed for the routes that sustain civilian and decentralised supply (Minten et al. 2023; Fig. 5c). The locally formed Yaw Defence Force illustrates the adaptive response, integrating Gangaw/Yaw’s surplus and routing it through mountain trails and secondary roads in a “horizontal-integration” model based on geographic connectivity and local self-sufficiency rather than drug economies.

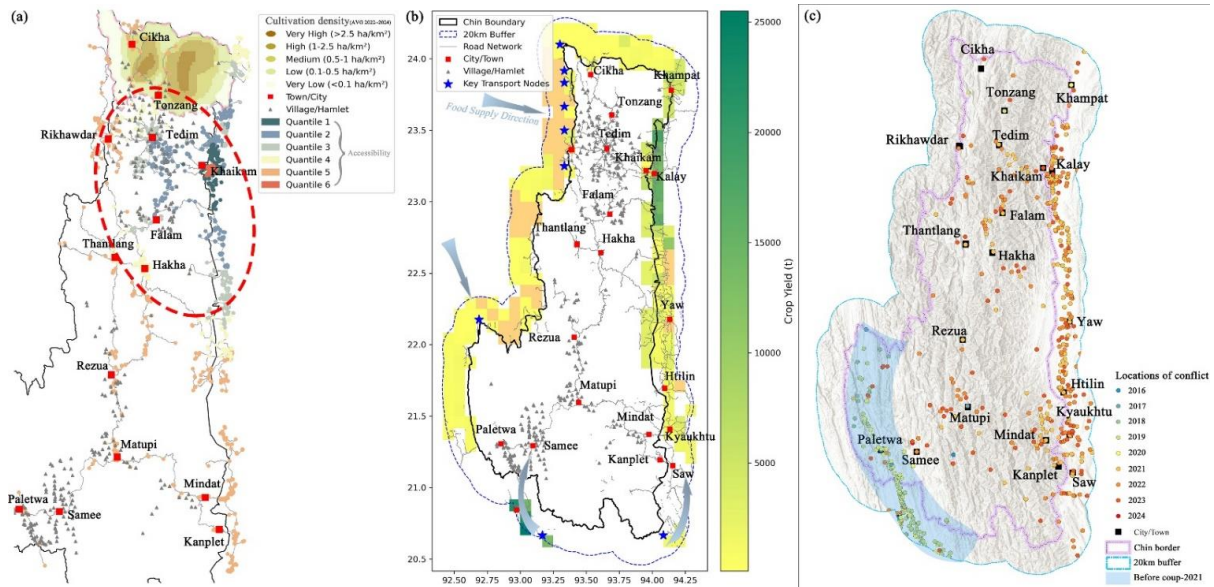


Fig. 5. Geospatial anatomy of food logistics in the western borderland. (a) Opium-poppy cultivation density (UNODC) and relative road accessibility (quantiles); (b) SPAM 2020 food-crop yield (10 km) with the OSM road network, the 20-km one-day-walk buffer, KD-tree transfer hubs (blue stars) and inferred food-supply direction; (c) ACLED conflict locations (2016–2024), with the pre-2021 southern cluster around Paletwa–Samee shaded.

5.3.2 Rikhawdar–Mizoram: the terminal cross-border gateway

When internal corridors are cut, external inflows become the binding constraint on highland access, and the Rikhawdar crossing into Mizoram becomes the critical gateway. Its low betweenness but recurrent appearance in displacement and resupply events marks it as a terminal node—an endpoint rather than a through-route. As airstrikes and checkpoint closures isolated towns such as Hakha and Thantlang, populations and trade shifted westward; Mizoram absorbed some 40,000 Chin refugees and channelled commodities and support back through Rikhawdar and adjacent tracks (Inkah 2024; Oxford Border Criminologies 2025). This interface is geopolitically contingent: India’s February 2024 suspension of the Free Movement Regime, the May 2024 closure of the Rikhawdar–Zokhawthar bridge, the December 2024 biometric-pass regime and ongoing border fencing have progressively hardened it (Mizzima 2024; Oxford Border Criminologies 2025). The redundancy it provides is therefore real but fragile, and decisive for the resilience of the whole system.

5.3.3 Paletwa and the Kaladan corridor: the strategic chokepoint

Paletwa is the system’s strategic chokepoint, where riverine circulation, territorial control and food access intersect, and it ranks first by betweenness in the regional network. As the inland river-port of the Kaladan corridor it commands the only substantial waterway linking southern Chin and Mizoram to the coast at Sittwe; the corridor’s internal structure—Paletwa to Kyauktaw, Minbya and Sittwe—appears directly in the network as the regional backbone. ACLED records show sustained AA activity around Paletwa predating the 2021 coup (Fig. 4c),

and in January 2024 the AA captured the town—the junta’s logistics base for the India and Bangladesh borders—complicating the Kaladan project and displacing over 50,000 people through subsequent air attacks (Radio Free Asia 2024; The Irrawaddy 2024). Because no comparable substitute exists for the Kaladan route, control of Paletwa plausibly determines whether the southern multimodal lifeline functions at all—a logistics-mediated mechanism consistent with the defining property of a chokepoint.

5.4 Corridor viability, robustness and systemic vulnerability (RQ2–RQ3)

Overlaying accessibility with annual conflict exposure shows corridor viability reorganising after 2021 (Fig. 5a, c): pre-coup conflict clustered in the south around Paletwa–Samee, while post-coup it diffused along the eastern transport spine and the Gangaw/Kalay approaches—where accessibility is highest and the production and highland corridors converge. The most accessible primary corridors are thus also the most exposed to interdiction, forcing adaptive rerouting onto lower-capacity redundancies (secondary roads, mountain trails, cross-border tracks) whose feasibility varies with season and conflict cycle.

The robustness simulation characterises how this dependence behaves under node loss (Fig. 6). The regional logistics network is highly tolerant of random loss—removing 5% of nodes at random leaves about 94% of the network connected, and 10% leaves 87%—but it is acutely vulnerable to targeted attack on high-betweenness nodes: removing the top 2% of nodes by betweenness collapses the largest connected component from 100% to 52%, the top 5% to 7%, and the top 10% to 2%. This robust-yet-fragile pattern (Albert et al. 2000) is consistent with a single-point-of-failure geography in the reconstructed network: connectivity depends on a thin backbone of high-betweenness nodes—above all Paletwa and the Kaladan corridor—so that an actor controlling a handful of them could in principle fragment circulation far out of proportion to the territory physically held. Systemic vulnerability is therefore concentrated at the non-redundant Paletwa–Kaladan chokepoint and along the high-exposure dry-zone approaches, while the Rikhawdar–Mizoram gateway supplies the principal—but geopolitically fragile—redundancy. This speaks to RQ3: in the reconstructed network, regional connectivity rests on very few links, indicating that their protection or substitution is a key lever for food access.

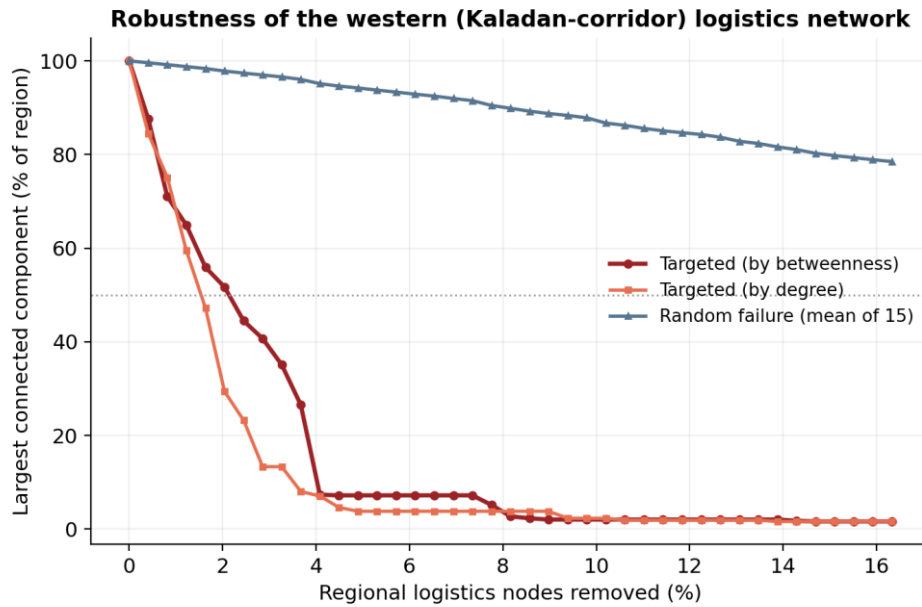


Fig. 6. Robustness of the western (Kaladan-corridor) logistics network under node removal. Largest connected component (% of region) as nodes are removed by betweenness (targeted), by degree, or at random (mean of 15 runs). Targeting the top 2% of nodes by betweenness collapses connectivity to ~52% and the top 5% to ~7%, whereas equivalent random loss leaves >90% intact.

6. Discussion

6.1 The chokepoint geography of conflict-induced hunger

The central finding is that food insecurity in the western Kaladan system is mediated by spatial circulation rather than by production shortfall alone. The Chin highlands are vulnerable because their access to staple food is filtered through three asymmetric interfaces: the dry-zone production belt, the Mizoram border gateway and the Kaladan riverine corridor. Each interface fails through a different mechanism—source-area erosion, border hardening or chokepoint capture—but their effects compound because the system has little redundancy. Paletwa therefore matters not simply as a town captured in conflict, but as the point where riverine mobility, borderland geopolitics and network brokerage converge. This is the empirical counterpart of the critical-logistics claim that power is exercised through the governance of flows and the holding of chokepoints (Cowen 2014; Danyluk 2023; Dittmer 2021), and it carries a corollary that an administrative reading would miss: the food security of the Chin highlands is governed by nodes that lie largely outside Chin State. The triad maps onto three distinct mechanisms of disruption—the erosion of the Gangaw/Yaw source, the hardening of the Rikhawdar gateway, and the capture of the Paletwa chokepoint—whose effects are not additive but compounding, since the loss of the Kaladan waterway throws load onto the Rikhawdar gateway while interdiction along the dry-zone approaches deepens dependence on both. Vulnerability, on this reading, is a property of the network rather than of any single

incident.

The broader theoretical implication is that conflict-induced food insecurity should be read as a problem of spatial circulation, not simply as a problem of production loss or conflict intensity. The western Kaladan system demonstrates how food access is filtered through a set of uneven interfaces: a dry-zone production belt that supplies surplus, a riverine corridor that concentrates movement, a cross-border gateway that provides external redundancy and a chokepoint that can translate local capture into system-wide fragmentation. This finding connects food-access theory with critical logistics and borderland geography. Entitlement and access approaches explain why food availability alone is insufficient; critical logistics explains why control over circulation becomes a political and military resource; borderland theory explains why Rikhawdar–Mizoram functions simultaneously as a lifeline, policy filter and potential closure point; and network vulnerability analysis provides the operational language for identifying non-substitutable nodes.

This also clarifies the applied contribution of the study. Conventional conflict maps identify where violence occurs, but they do not necessarily show where the failure of circulation will have the largest food-access consequences. Conversely, food-security assessments identify populations at risk, but often do not locate the nodes and corridors through which that risk is produced. The node–corridor–chokepoint framework links these two forms of evidence. It identifies where pre-positioning may reduce last-mile isolation, where negotiated access would have the greatest systemic value, where cross-border coordination is most consequential and where alternative routes would need to be strengthened before primary corridors fail. In the western Kaladan case, this means treating Paletwa–Kaladan, Gangaw/Kalay and Rikhawdar–Mizoram not as isolated sites but as coupled leverage points in a single functional food-supply system.

6.2 Implications for humanitarian logistics and food-access assessment

These findings carry direct operational value, which is where the framework speaks most strongly to applied geography. The robustness analysis identifies, in priority order, the nodes whose failure cascades most widely—the non-redundant Paletwa–Kaladan chokepoint and the high-exposure dry-zone approaches—allowing humanitarian actors to focus contingency planning, pre-positioning and negotiated-access efforts where they matter most rather than spreading attention evenly across a conflict map. The cross-border analysis shows that resilience for the Chin highlands depends substantially on the Rikhawdar–Mizoram gateway and the Indian border policy governing it, making cross-border coordination a humanitarian-

logistics priority, especially given the 2024–2025 hardening of that interface. The one-day-walk transfer radius offers a tractable last-mile heuristic where motorised access is denied. Set against UN findings that the blockade of aid to western Myanmar likely constitutes a war crime and that famine looms in Rakhine (OHCHR 2025), mapping these leverage points is not merely analytical but protection-relevant.

6.3 Methodological contribution and transferability

Methodologically, the study demonstrates that unstructured conflict narratives can be fused with spatial production and accessibility data, and then subjected to network and robustness analysis, to identify vulnerable supply geographies that no single source reveals. Defining the study unit functionally—and benchmarking the regional network against the national one—lets the analysis capture a food system that crosses administrative and international boundaries. The pipeline relies on globally available inputs (independent news, ACLED, OSM, SPAM) and standard network methods, so it is transferable to other contested borderlands where food access turns on the control of a few nodes and corridors (Pech et al. 2021; Zhukov 2012). It is also extensible: the same event base supports actor–place control networks, named-corridor disruption layers, least-cost accessibility modelling, population-cut-off estimation and market-price coupling.

6.4 Limitations

Several limitations qualify the results. The reliance on a single, English-language independent aggregator introduces reporting and linguistic biases that may under-represent thinly covered areas (Baum and Zhukov 2015; Shaver et al. 2022), and LLM-based extraction can carry geographic biases (Manvi et al. 2024); neither substitutes for field verification. The current validation is an event-table audit and external-validation design rather than a completed hand-coded P/R/F1 assessment. Network edges denote co-reporting rather than verified physical links, so the network is a reported-association proxy for the supply system; betweenness reflects realised co-occurrence in reported events, shaped by what is reported as well as by what occurs, and the boundary of a functional region is itself a modelling choice that affects which nodes appear central. A transport-constrained reconstruction using OSM road/river adjacency would tighten this correspondence. The accessibility analysis uses a relative travel-cost and buffer approximation rather than fully calibrated least-cost transport modelling, and SPAM production is a modelled 10-km surface. Finally, the findings are a temporally specific snapshot of a fast-moving conflict; multilingual and multi-source corpora, household food-security data, ground-truthed transport costs and an independently coded

validation subset would strengthen both validity and operational usefulness.

7. Conclusion

This paper began from a corridor and the system of provisioning it sustains, and developed an applied geospatial framework for mapping food-logistics nodes and adaptive supply corridors under conflict blockade. Operationalised for the functional food-supply system of Myanmar's western borderland with 11,644 text-derived, geocoded events for 2016–2026, fused with conflict, transport and crop-production data, the framework renders visible a geography that neither a single dataset nor an administrative map discloses.

The three questions are answered in turn. Food access appears to be organised by a triadic geography (RQ1) of functionally distinct nodes—Gangaw/Yaw and the dry-zone belt as a production source, Rikhawdar–Mizoram as a terminal cross-border gateway, and Paletwa with the Kaladan corridor as the strategic chokepoint, which ranks first by betweenness within the regional network and twentieth of 2,793 nodes nationally. Conflict appears to reconfigure corridor viability by displacing interdiction onto the most accessible trunk routes and forcing adaptive rerouting onto lower-capacity alternatives whose feasibility shifts with season and conflict cycle (RQ2). And the reconstructed network indicates an acute, asymmetric vulnerability (RQ3): tolerant of random loss, it fragments once a handful of high-betweenness chokepoints are removed—the top 2% taking connectivity from 100% to 52%, the top 5% to 7%—so that resilience rests on a thin backbone whose principal redundancy, the Rikhawdar–Mizoram gateway, has itself been hardened since 2024.

The contribution is, finally, twofold. Substantively, the paper offers reproducible network and robustness evidence consistent with a reading in which conflict-induced hunger in contested borderlands is mediated by the capture of a few critical nodes and corridors—nodes that may lie beyond the administrative unit that bears the consequences—and it maps those candidate leverage points explicitly. Methodologically, it shows that unstructured conflict narratives can be fused with spatial production and accessibility data and then subjected to network and robustness analysis, yielding a workflow that travels to other contested borderlands where food access turns on the control of a few passages. As blockade continues to convert geopolitical violence into localised hunger, the task of locating where circulation can be sustained, and where it can be severed, is not merely analytical but a precondition for protecting it.

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